Deven Desai

devendesai116@gmail.com | San Jose, CA | +1(669)-292-6556 | LinkedIn | GitHub

SUMMARY

Software Engineering Master's student and Stanford Hackathon winner specializing in full stack software development, agentic workflow engines, and developer productivity tooling.

PROJECTS

AgenticFlow | Full Stack Workflow Automation & Developer Platform GitHub

- Automated schema transformations and data migrations across cloud systems by architecting an autonomous workflow execution engine using TypeScript, Node.js, Python, and LangGraph.
- Increased service execution throughput by 40 percent by designing concurrent reasoning loops and gRPC microservice interfaces as measured by system execution traces.
- Ensured high data trust and enterprise privacy across platform boundaries by engineering an automated validation layer verified via continuous unit test suites.

OmniTrack AI | Multimodal Processing & AI Agent Evaluation Engine GitHub

- Processed unstructured streaming inputs and user metadata natively in cloud infrastructure by engineering a zero to one full stack data application using React, TypeScript, and Python.
- Achieved sub 100ms vector search latency by independently optimizing Qdrant indexing algorithms to deliver precise context without prompt bloat.
- Enhanced output reliability and developer feedback loops by 30 percent by building continuous evaluation pipelines, guardrails, and automated verification suites.

Low Latency Search Engine | High Performance Information Discovery Platform GitHub

- Analyzed and queried high volume structured datasets in real time by engineering a custom full stack analytical search interface as a solo developer using React and Java.
- Decreased search query response latency by 35 percent by developing optimized retrieval algorithms as measured by end to end response times.
- Optimized execution throughput by 20 percent by tuning underlying data structures and batch data execution routines.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL (Postgres), Go, C++, Java, HTML5 CSS3
Full Stack & AI Tooling: React, Redux, Node.js, Express, RESTful APIs, gRPC Microservices, Agentic Workflows (LangGraph)
Data & Cloud Infrastructure: PostgreSQL (Aurora), Redis (ElastiCache), OpenSearch, Relational Schema Design, Async Data Pipelines
Developer Tools & Practices: Git, GitHub, Playwright, CI CD, Infrastructure automation, Automated Testing, System Architecture

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.